



Take-Home Assignment

Backend & Data Engineer

At a glance

- **Effort:** ~3–4 hours of building + ~30 min to record a short Loom
- **Deadline:** 7 days
- **Submit:** a GitHub repo link + a Loom link (reply to the email this came in)

We run telematics for 2 lakh+ tractors. The data is never clean, and small teams here own things end-to-end. This is a small slice of real work at Carnot. **AI tools are allowed and expected** — we use them daily. The Loom is where you show us the thinking behind the code.

The Task

You get two files:

- **pings.csv** — raw device pings: device_id, ts, ignition, odometer_km
- **vehicles.csv** — the vehicle master: device_id, registration_no, model, region, owner_type

Build a small thing that:

1. Computes, per vehicle, for the data period: total_distance_km, active_days (days it actually moved), and a status of active / inactive / no_data.

*Distance comes from the odometer: **sum the increases between consecutive readings in time order**. A decrease means the device was reset/replaced — don't count that as travel. active = moved in the last 7 days of the period.*

2. **Serves it** via one endpoint: GET /vehicles/{device_id}/usage → that vehicle's computed fields.

The data is realistically messy — handle the issues that would otherwise corrupt your numbers, and note briefly what you found. **Don't over-engineer:** no DB, auth, or frontend needed; a file or in-memory store is fine.

Deliverables

- The code, runnable from a short **README** (how to run it + a couple of lines on the data issues you hit and how you used AI tools).
- A test or two on the parts that are easy to get wrong. Not full coverage.
- A **Loom (5–10 min)**: screen-share your code and talk us through (a) how you separated concerns and why, (b) your 2–3 key decisions, and (c) if this ran daily in production, how would you run it, and what's the first thing you'd change if it had to handle 100M pings a day?

On the Loom

- Rough and unscripted is exactly what we want — **no webcam, just screen + voice, a first take is fine.**
- **We won't review a submission without it.**

What we're NOT evaluating

Relax about these

- Whether you got the “right” output — there isn't one; your judgement calls matter more.
- How fast you finished, or how polished it looks.

If this was fun to work on, that's a good sign — for you and for us.